Fundamental subspaces

Let us start with the following system of equations

$$Ax = b: \quad \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

Suppose A is fixed. There are two ways in which we can study the above

- b is given and we are asked to find x such that $Ax = b$. The solution or vector x may or may not exist, and if it exists, may or may not be unique. That is, once b is given, the solution set may be empty or non-empty.
- Conversely, we require the solution set to be empty or non-empty. In this case, we are asked to find all such $b's$ such that the solution set of $Ax = b$ is empty or non-empty.

We have discussed the first case in Module 1. We discuss the second case now.

Let the columns of the matrix A be denoted by c_j, that is

$$c_1 = \begin{pmatrix} a_{11} \\ a_{21} \\ \cdots \\ a_{m1} \end{pmatrix}, \quad c_1 = \begin{pmatrix} a_{12} \\ a_{22} \\ \cdots \\ a_{m2} \end{pmatrix}, \cdots, c_n = \begin{pmatrix} a_{1n} \\ a_{2n} \\ \cdots \\ a_{mn} \end{pmatrix}$$

Then each c_j is a vector in $\mathbb{R}^m$ (*where m is the number of rows in A*), and there are n such $c_j's$ (*where n is the number of columns in A*). If we write $Ax = b$ explicitly, we get

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

which can be rearranged and written as

$$x_1 \begin{pmatrix} a_{11} \\ a_{21} \\ \cdots \\ a_{m1} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ a_{22} \\ \cdots \\ a_{m2} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ \cdots \\ a_{mn} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \cdots \\ b_m \end{pmatrix} \Rightarrow x_1c_1 + x_2c_2 + \cdots + x_nc_n = b$$

What the above expression means is that the constant vector b can also be expressed as a linear combination of the column vectors of A.

Thus asking to solve a system $Ax = b$ is equivalent to asking whether the vector b can be expressed as a linear combination of the column vectors of A. If it is possible, the coefficients $x_j's$ that we obtain from such a linear combination is the solution to the system $Ax = b$. If not, then the system does not have a solution.

0.1 Column space

The above discussion shows that linear combinations of column vectors of A are important when discussing existence of solutions. We give a name to the set of all such linear combinations.

Definition of column space

Definition 1. The column space $\mathcal{C}(A)$ of a $m \times n$ matrix A is the set of linear combinations of the column vectors $c_1, c_2, \cdots, c_n$ of A. In terms of the concept of span, the column space is the space spanned by the column vectors of A.

- **Where does this space lie?** As discussed above, each column c_j contains m coordinates and hence is a vector in $\mathbb{R}^m$. Thus, the vectors formed by their linear combinations and hence the column space of A will also lie in $\mathbb{R}^m$.
- **What does b got to do with the column space?** If the system $Ax = b$ has to have a solution, we should be able to express b as a linear combination of the columns of A. In other words, b has to lie in the column space of A. Precisely, the column space of A is the space of all vectors b in $\mathbb{R}^m$ for which the system $Ax = b$ is solvable.
- **When is $Ax = b$ not solvable?** If b lies in the column space of A, we will always get a solution, unique or infinite. Hence, the system does not have a solution for all such vectors b in $\mathbb{R}^m$ which do not lie in the column space of A. *In such cases, that is we need a solution of $Ax = b$ when it is not solvable, we use the method of **least square**, which is discussed in Module 5.*

0.2 Row space

Let A be an $m \times n$ matrix and consider the rows of A. Since there are n columns, each row of A can be considered as a vector in $\mathbb{R}^n$. The set of all linear combinations of the row vectors is called the row space of A and is denoted by $\text{Rowsp}(A)$.

Definition of row space

Definition 2. The row space $\text{Rowsp}(A)$ of a $m \times n$ matrix A is the set of linear combinations of the row vectors $r_1, r_2, \cdots, r_m$ of A. In terms of the concept of span, the row space is the space spanned by the row vectors of A.

For any $m \times n$ matrix A, we understand the solution set of $Ax = b$ via the column space of A. Since the rows of A are the columns of A^T, $\text{Rowsp}(A)$ is used to understand the solution set of the system $A^T y = c$. This is because to have a solution c, we should be able to express c as a linear combination of the columns of A^T, that is as a linear combination of the rows of A.

Without going into details, the row space will lie in $\mathbb{R}^n$ and we have

$$\text{Colsp}(A) = \text{Rowsp}(A^T) \text{ and } \text{Rowsp}(A) = \text{Colsp}(A^T).$$

0.3 Null space

Recall the example of the solution set of a homogeneous system of equation $Ax = \Theta$. We give a name to this solution space.

Definition 3. The solution set of the homogeneous system of equation $Ax = \Theta$ is called the null space of A and denoted by $\text{Nullsp}(A)$.

Definition of null space

To verify if a given vector v lies in the null space of A, just compute Av to see if the result is the zero vector.

Let us look at some examples on column space, row space and null space. The basic idea in these examples is to find a basis for the above three subspaces. Note that, even is we define column space of A as a linear combination of columns of A, it is possible that some of the columns are redundant, that is they are themselves linear combination of remaining columns of A and hence can be discarded. Same goes for row space.

Thus, in these examples, we will see *how to find the columns and rows that are linearly independent and hence form a basis for the column space and row space respectively*. Similarly, we will discuss how to find a basis for the null space also.

Example 1. Let us find bases for the row space, column space, and the null space for the following matrix

$$A = \begin{pmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{pmatrix}$$

The first step is to reduce the given matrix to row reduced echelon form. For A we get

$$B = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

- For row space, we use the result that the non-zero rows of the reduced echelon form B of any matrix A forms a basis for B as well as A. Hence, a basis for the row space of the given matrix A is

$$\mathcal{B}_{rowsp} = \{(1, 0, 1, 0, 1), (0, 1, -2, 0, 3), (0, 0, 0, 1, -5)\}$$

- For column space, we look at the columns of the row reduced echelon form that contain the pivot. In the present case, columns 1,2 and 4 of B contain the pivot. Hence columns 1,2 and 4 of A forms the basis for the column space of A. Thus

$$\mathcal{B}_{colsp} = \left\{ \begin{pmatrix} -2 \\ 1 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} -5 \\ 3 \\ 11 \\ 7 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 7 \\ 5 \end{pmatrix} \right\}$$

- For null space, we write the corresponding equations from the row reduced echelon form

$$\begin{aligned} x_1 + x_3 + x_5 &= 0 \\ x_2 - 2x_3 + 3x_5 &= 0 \\ x_4 - 5x_5 &= 0 \end{aligned}$$

Consider $x_5 = s$ as the free variable. From the last equation$x_4 = 5s$. For the second equation, consider $x_3 = t$ as the second free variable. Then $x_2 = 2t - 3s$. From the first equation, $x_1 = -s - t$. Hence the null space in parametric form is

$$\begin{aligned} \text{Nullsp}(A) &= \{(-s - t, 2t - 3s, t, 5s, s) | (s, t) \in \mathbb{R}^2\} \\ &= \{s(-1, -3, 0, 5, 1) + t(-1, 2, 1, 0, 0) | (s, t) \in \mathbb{R}^2\} \end{aligned}$$

This gives the basis for the null space as

$$\mathcal{B}_{nullsp} = \left\{ \begin{pmatrix} -1 \\ -3 \\ 0 \\ 5 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

Example 2. Let us find a basis for the null space of the matrix

$$A = \begin{pmatrix} -3 & 6 & -1 & 1 & 7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{pmatrix}$$

The first step is to reduce A to the row reduced echelon form

$$B = \begin{pmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The corresponding system of equations is

$$\begin{aligned} x_1 - 2x_2 - x_4 + 3x_5 &= 0 \\ x_3 + 2x_4 - 2x_5 &= 0 \end{aligned}$$

Let $x_4 = s$ and $x_5 = t$ be the free variables. Then $x_3 = -2s + 2t$. For the first equation, we need another free variable, let $x_2 = r$ so that $x_1 = 2r + s - 3t$. The solution set is

$$\begin{aligned} \text{Nullsp}(A) &= \{(2r + s - 3t, r, -2s + 2t, s, t) | (r, s, t) \in \mathbb{R}^3\} \\ &= \{r(2, 1, 0, 0, 0) + s(1, 0, -2, 1, 0) + t(-3, 0, 2, 0, 1) | (r, s, t) \in \mathbb{R}^3\} \end{aligned}$$

This gives a basis for the nullspace of A as

$$\mathcal{B}_{nullsp} = \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Example 3. Let

$$A = \begin{pmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{pmatrix} \text{ and } b = \begin{pmatrix} 3 \\ 3 \\ -4 \end{pmatrix}$$

We need to determine whether b lies in the column space of A. There are two ways in which we can answer.

- The first method is to use the definition of linear combination to try and write b as a linear combination of the columns of A. Thus, let

  $$(3, 3, -4) = \alpha_1(1, -4, -3) + \alpha_2(-3, 6, 7) + \alpha_3(-4, -2, 6)$$

  which gives the system of equations

  $$\begin{aligned} \alpha_1 - 3\alpha_2 - 4\alpha_3 &= 3 \\ -4\alpha_1 + 6\alpha_2 - 2\alpha_3 &= 3 \\ -3\alpha_1 + 7\alpha_2 + 6\alpha_3 &= -4 \end{aligned}$$

  If a solution exists, then b lies in the column space of A. If no solution exists, then b does not lie in the column space of A.

- The second method is based on the fact b lies in the column space of A if and only if $Ax = b$ has a solution. Hence we use the method of augmented matrix

  $$\begin{pmatrix} 1 & -3 & -4 & 3 \\ -4 & 6 & -2 & 3 \\ -3 & 7 & 6 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & -2 & -6 & 5 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  to determine if it has a solution. If a solution exists, then b lies in the column space of A. If no solution exists, then b does not lie in the column space of A.

Example 4. Find a basis for the vector space V spanned by the vectors $v_1 = (1, 1, 0)$, $v_2 = (0, 1, 1)$, $v_3 = (2, 3, 1)$ and $v_4 = (1, 1, 1)$.

Since the underlying vector space V is $\mathbb{R}^3$, it is of dimension 3 and any basis will contain exactly 3 vectors. Thus the set $\{v_1, v_2, v_3, v_4\}$ is certainly LD. We need to first identify the one that is redundant and remove it. The linear combination method will not help as it just tells us whether the vectors are LD or LI. But the method of finding a basis for column space will work. Write the given vectors as the columns of a matrix, transform it to row-reduced echelon form and identify the LI columns. Those are the required Li vectors.

$$\begin{pmatrix} 1 & 0 & 2 & 1 \\ 1 & 1 & 3 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Columns 1,2 and 4 contain the pivot and hence the vectors v_1, v_2, v_4 is LI. Now prove that v_1, v_2, v_4 span $V = \mathbb{R}^3$.

Exercises

1. Determine the dimension as well as a basis for the space spanned by

$$S = \{(1, 2, 1), (1, 0, 2), (5, 6, 7)\}$$

(*S* is LD, $\{(1, 2, 1), (1, 0, 2)\}$)

2. Determine the dimension as well as a basis for the space spanned by

$$S = \{(1, 2, -1, 3), (1, 0, 0, 2), (2, 8, -4, 8), (1, 1, 1, 1), (3, 3, 0, 6)\}$$

(*S* is LD, $\{(1, 2, -1, 3), (0, -2, 1, -1), (0, 0, 3, -3)\}$)

3. Consider the following matrix and the column vector

$$A = \begin{pmatrix} 1 & 2 & 2 & 0 & 5 \\ 2 & 4 & 3 & 1 & 8 \\ 3 & 6 & 1 & 5 & 5 \end{pmatrix}, \qquad v = \begin{pmatrix} -8 \\ 1 \\ 3 \\ 3 \\ 0 \end{pmatrix}$$

Determine if v belongs to the null space of A by the following methods.

- Direct verification. (YES)
- Finding a basis for the null space of A and verifying if v can be expressed as a linear combination of the basis vectors.
  (A basis is $\{(-2, 1, 0, 0, 0), (-2, 0, 1, 1, 0), (-1, 0, -2, 0, 1)\}$)

Does v belong to the column space of A? (NO)